

Example 1

If a total of 33 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses two 25 kHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell if a system uses (a) four-cell reuse, (b) seven-cell reuse, and (c) 12-cell reuse. If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems.

Solution:

Given: Total bandwidth = 33 MHz

Channel bandwidth = $25 \text{ kHz} \times 2 \text{ simplex channels} = 50 \text{ kHz/duplex channel}$

Total available channels = $33,000/50 = 660$ channels

(a) For $N = 4$, total number of channels available per cell = $660/4 \approx 165$ channels.

(b) For $N = 7$, total number of channels available per cell = $660/7 \approx 95$ channels.

(c) For $N = 12$, total number of channels available per cell = $660/12 \approx 55$ channels.

A 1 MHz spectrum for control channels implies that there are $1000/50 = 20$ control channels out of the 660 channels available. To evenly distribute the control and voice channels, simply allocate the same number of voice channels in each cell wherever possible. Here, the 660 channels must be evenly distributed to each cell within the cluster. In practice, only the 640 voice channels would be allocated, since the control channels are allocated separately as 1 per cell.

(a) For $N = 4$, we can have five control channels and 160 voice channels per cell. In practice, however, each cell only needs a single control channel (the control channels have a greater reuse distance than the voice channels).

Thus, one control channel and 160 voice channels would be assigned to each cell.

(b) For $N = 7$, four cells with three control channels and 92 voice channels, two cells with three control channels and 90 voice channels, and one cell with two control channels and 92 voice channels could be allocated. In practice, however, each cell would have one control channel, four cells would have 91 voice channels, and three cells would have 92 voice channels.

(c) For $N = 12$, we can have eight cells with two control channels and 53 voice channels, and four cells with one control channel and 54 voice channels each. In an actual system, each cell would have one control channel, eight cells would have 53 voice channels, and four cells would have 54 voice channels.

Question1

We consider a cellular system in which total available voice channels to handle the traffic are 960. The area of each cell is 6 km^2 and the total coverage area of the system is 2000 km^2 . Calculate (a) the system capacity if the cluster size, N (reuse factor), is 4 and (b) the system capacity if the cluster size is 7. How many times would a cluster of size 4 have to be replicated to cover the entire cellular area? Does decreasing the reuse factor N increase the system capacity? Explain.

- Total available channels = 960
- Cell area = 6 km^2
- Total coverage area = 2000 km^2
 $N = 4$
- Area of a cluster with reuse $N = 4$: $4 \times 6 = 24 \text{ km}^2$
- Number of clusters for covering total area with N equals 4 = $2000/24 = 83.33 \sim 83$
- Number of channels per cell = $960/4 = 240$
- System capacity = $83 \times 960 = 79,680$ channels

- Area of cluster with reuse $N = 7$: $7 \times 6 = 42 \text{ km}^2$
- Number of clusters for covering total area with N equals $7 = 2000/42 = 47.62 \sim 48$
- Number of channels per cell $= 960/7 = 137.15 \sim 137$
- System capacity $= 48 \times 960 = 46,080$ channels

It is evident when we decrease the value of N from 7 to 4, we increase the system capacity from 46,080 to 79,680 channels. Thus, decreasing the reuse factor (N) increases the system capacity.

Example: If a signal to interference ratio of 15 dB is required for satisfactory forward channel performance of a cellular system, what is the frequency reuse factor and cluster size that should be used for maximum capacity if the path loss exponent is (a) $n = 4$, (b) $n = 3$? Assume that there are 6 co-channels cells in the first tier, and all of them are at the same distance from the mobile. Use suitable approximations.

(a) $n = 4$

First, let us consider a 7-cell reuse pattern.

the co-channel reuse ratio $D/R = \sqrt{3N} = 4.583$.

the signal-to-noise interference ratio is given by

$$S/i = (1/6) \times (4.583)^4 = 75.3 = 18.66 \text{ dB.}$$

Since this is greater than the minimum required S/I , $N = 7$ can be used.

b) $n = 3$

First, let us consider a 7-cell reuse pattern.

the signal-to-interference ratio is given by

$$S/I = (1/6) \times (4.583)^3 = 16.04 = 12.05 \text{ dB.}$$

Since this is less than the minimum required S/I , we need to use a larger N .

The next possible value of N is 12, ($i = j = 2$).

The corresponding co-channel ratio is given by as $D/R = \sqrt{3N} = 6.0$.

$$S/I = (1/6) \times 6^3 = 36 = 15.56 \text{ dB.}$$

Since this is greater than the minimum required S/I , $N = 12$ can be used.